

FedXDDoS: An Explainable Federated Learning Framework for DDoS Attack Detection

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Abstract—Distributed Denial of Service (DDoS) attacks have emerged as a significant security challenge in software-defined networking (SDN) environments, where surges in malicious traffic can disrupt network operations and degrade service availability. Although deep learning-based intrusion detection systems have achieved excellent performance in identifying malicious activities, many existing solutions depend on centralized training and lack interpretability. To overcome these challenges, This study introduces FedXDDoS, an explainable federated learning framework for privacy-preserving DDoS attack detection and explanation in distributed SDN environments. This framework utilizes multiple deep learning architectures under both FedAvg and FedProx optimization strategies to facilitate collaborative model training with no sharing of raw network traffic data. To demonstrate the effectiveness of the proposed approach, which achieves a best accuracy of 99.69% and an F1 score of 99.67%, the experiments were conducted on the DDoS-SDN dataset using four federated clients. Additionally, LIME-based explainability is integrated to enhance transparency and trustworthiness by offering feature-level interpretations of model predictions. The results show that FedXDDoS provides an accurate and interpretable approach for detecting DDoS attacks in contemporary SDN environments.

Index Terms—DDoS Detection, Federated Learning, Explainable Artificial Intelligence, Deep Learning, Software-Defined Networking, Network Security.

I. INTRODUCTION

Software-Defined Networking (SDN), the Internet of Things (IoT), and Cloud Computing (CC) have become pivotal technologies in modern communication infrastructures because they enable centralized control and efficient resource utilization management [1], [2]. However, this reliance on interconnected systems has also heightened the complexity of cyber threats, particularly Distributed Denial of Service (DDoS) attacks [3]. These attacks produce massive volumes of malicious traffic that can deplete network resources and disrupt

normal services, underscoring the critical need for precise, real-time DDoS detection in SDN environments.

Machine learning (ML) and deep learning (DL)-based detection systems have demonstrated potential performance in identifying and classifying malicious traffic patterns [4]. Several DL techniques have been widely incorporated due to their capability to learn complex and malicious traffic characteristics and improve classification accuracy. On the other hand, most current methods depend on centralized training systems that need to share a lot of traffic data with a central server, which brings up issues about privacy, how well they can scale, and the amount of data being sent back and forth [5]. Federated Learning (FL) has become a new approach that helps protect privacy by letting multiple clients work together to train a shared model without sending their raw data to a central place [6], [7]. However, traditional FL methods like Federated Averaging (FedAvg) often have problems with training stability when dealing with different types of data that aren’t the same across clients [8], [9]. To fix this, Federated Proximal (FedProx) uses a special kind of regularization that helps the training process stay stable even when the data is varied [9]. Additionally, many machine learning-based intrusion detection systems are like black boxes, making it hard for security experts to understand or trust the results they produce [10]. Recent studies have shown the effectiveness of Explainable Artificial Intelligence (XAI) in improving the transparency and explainability of ML models across various application domains [11]–[13]. Despite recent advances in FL-based intrusion detection, the integration of explainability with privacy-preserving collaborative learning remains largely unexplored for DDoS detection in SDN environments. Motivated by these challenges, this study proposes FedXDDoS, an explainable FL framework for privacy-preserving DDoS attack detection in distributed SDN

environments. The proposed framework investigates multiple DL architectures under both FedAvg and FedProx optimization strategies to facilitate collaborative learning without exposing raw network traffic data. In addition, LIME-based explainability is incorporated to provide feature-level interpretations of model predictions and enhance transparency.

This study aims to develop a robust, privacy-preserving, and interpretable DDoS attack detection framework for distributed SDN environments while addressing the limitations of centralized and black-box intrusion detection approaches. The major contributions of this study are summarized as follows:

- An explainable federated learning framework, FedXDDoS, is developed for DDoS attack detection in distributed SDN environments, where raw network traffic remains at individual clients.
- A systematic comparative evaluation of five deep learning architectures, namely MLP, 1D-CNN, LSTM, GRU, and BiLSTM, is conducted under both FedAvg and FedProx within a common federated DDoS detection framework.
- LIME-based feature-level explanations are integrated with the federated global model to analyze the traffic characteristics contributing to normal and DDoS predictions, providing an interpretable view of model decisions.
- The framework jointly analyzes classification performance, convergence behavior, and model interpretability, providing an integrated evaluation of federated DDoS detection in distributed SDN environments.

The rest of this paper is organized like this. Section II looks at previous research on using ML, DL, and FL for detecting DDoS attacks. Section III introduces the FedXDDoS framework we created. Section IV covers the experiments we ran and the performance of our method. Lastly, Section V wraps up the paper and suggests ideas for future work.

II. RELATED WORKS

The increasing vulnerability of SDN environments to DDoS attacks has encouraged researchers to adopt ML, DL, and FL-based detection frameworks. Recent studies demonstrate that intelligent learning models can effectively detect malicious network traffic patterns and improve network security.

Sawah *et al.* [14] proposed a Random Forest-based DDoS detection framework integrated with Backward Elimination and Grid Search optimization using the DDoS-SDN dataset. Similarly, Banju *et al.* [15] employed the K-Nearest Neighbor (KNN) algorithm for SDN-based DDoS classification, while Haque *et al.* [16] reported superior performance using XG-Boost and Random Forest classifiers. Butt *et al.* [17] further introduced an ensemble learning framework with dynamic feature selection to improve detection reliability. Although these approaches achieved high classification accuracy, they mainly relied on centralized architectures and provided limited interpretability. To address privacy concerns associated with centralized learning, FL has emerged as a promising approach for collaborative intrusion detection. FLAD proposed by Doriguzzi-Corin *et al.* [18] introduced an adaptive FL

mechanism for DDoS detection under heterogeneous environments. Likewise, Li *et al.* [19] developed a federated learning empowered architecture for Industrial IoT environments using GRU-based models. More recently, Doriguzzi-Corin *et al.* [20] analyzed multiple FL algorithms under non-IID settings and highlighted convergence instability in conventional FedAvg approaches. Similarly, Shirvani *et al.* [21] investigated federated learning for IoT security using deep autoencoders and aggregation strategies such as FedAvg and FedAvgM. Despite their effectiveness, explainability and interpretability remained largely unexplored.

Although significant progress has been achieved in ML, DL, and FL-based DDoS detection, several challenges remain unresolved. Most existing methods either depend on centralized architectures or lack transparency in model decisions. These limitations motivate the development of an explainable FL framework capable of providing robust DDoS detection with improved explainability in distributed SDN environments.

III. METHODOLOGY

A. Proposed FedXDDoS Framework

The architecture of the proposed FedXDDoS framework is illustrated in Fig. 1. The framework follows a client-server FL paradigm in which multiple clients collaboratively train a global DDoS detection model while keeping their local network traffic data on the client side. The methodology is organized into the proposed framework, dataset description, data preprocessing and federated partitioning, explainability, and experimental configuration to provide a systematic description of the proposed approach. This approach enables collaborative model training without exposing sensitive traffic records to the central server. Initially, the DDoS dataset is preprocessed through data cleaning, feature normalization, and label encoding. The processed data are then distributed among K clients, where each client k possesses a local dataset D_k . During training, all data remain on the client side, thereby preserving data privacy throughout the learning process.

At communication round t , the server initializes and broadcasts the global model parameters w^t to all participating clients. Depending on the experimental configuration, the global model can be implemented using MLP, 1D-CNN, GRU, LSTM, or BiLSTM architectures. Once the global model is distributed, each client performs local training on its private dataset for E local epochs. The local optimization objective at client k is defined as:

$$F_k(w) = \frac{1}{|D_k|} \sum_{i \in D_k} l(w; x_i, y_i), \quad (1)$$

where $l(\cdot)$ denotes the loss function, w represents the model parameters, and (x_i, y_i) corresponds to the i^{th} training sample. Each client minimizes the local objective in Eq. (1) to obtain updated model parameters. After local training, only the updated model weights are transmitted to the server, while the underlying training data remain private.

TABLE I
QUALITATIVE COMPARISON OF EXISTING DDoS DETECTION APPROACHES

Study	Dataset	Best Performing Model	Learning Setting	Limitation
Banju <i>et al.</i> [15]	DDoS-SDN	KNN	Centralized	Scalability remains challenging for large-scale traffic
Haque <i>et al.</i> [16]	DDoS-SDN	XGBoost	Centralized	Distributed privacy protection was not considered
Butt <i>et al.</i> [17]	DDoS-SDN	Ensemble Learning	Centralized	Model interpretability remained limited
Doriguzzi <i>et al.</i> [18]	DDoS Dataset	FLAD	Federated	Explainability mechanisms were not incorporated
Li <i>et al.</i> [19]	IIoT Traffic Dataset	GRU-based FL	Federated	Computational overhead increased in IIoT settings
Doriguzzi <i>et al.</i> [20]	CIC-DDoS2019	Comparative FL Models	Federated	Convergence instability under non-IID environments
Shirvani <i>et al.</i> [21]	N-BaIoT	FedAvgM and Autoencoder	Federated	Explainability aspects remained unexplored

The server aggregates the received local models using the selected federated optimization strategy. For FedAvg, the global model is updated through weighted averaging of the client parameters as follows:

$$w^{t+1} = \sum_{k=1}^K \frac{n_k}{n} w_k^{t+1}, \quad (2)$$

where n_k denotes the number of samples at client k , $n = \sum_{k=1}^K n_k$, and K represents the total number of participating clients. Thus, clients with larger datasets contribute proportionally more to the global model update.

To address data heterogeneity across clients, FedProx incorporates a proximal regularization term into the local optimization objective. The modified objective function is expressed as

$$F_k(w) = f_k(w) + \frac{\mu}{2} \|w - w^t\|^2, \quad (3)$$

where $f_k(w)$ represents the original local loss function, w^t denotes the current global model parameters, and μ controls the strength of the proximal regularization term. The proximal term constrains excessive deviation of local model parameters from the global model, thereby improving training stability under heterogeneous (non-IID) data distributions.

After local optimization and aggregation, the server generates an updated global model and redistributes it to all participating clients. The clients then perform another round of local training using the updated parameters. This communication and aggregation process is repeated for the predefined number of communication rounds. Finally, LIME is applied to the trained global model to provide feature-level interpretations of its predictions.

B. Dataset Description

The experimental evaluation is carried out using the publicly available DDoS-SDN dataset obtained from Kaggle [22]. The

dataset contains 104,345 network traffic instances collected from an SDN environment, comprising 22 traffic-related features and one binary target label. The features capture network-flow characteristics such as packet count, packet rate, byte count, duration, and protocol information. The target label distinguishes normal traffic (0) from DDoS attack traffic (1).

C. Data Preprocessing

The dataset is preprocessed by removing records with missing values, encoding categorical features and labels using LabelEncoder, and applying a stratified 80/20 train-test split. Finally, StandardScaler is used to normalize feature values, improving training stability and convergence.

D. Explainable AI

To improve the transparency of the proposed FedXDDoS framework, LIME [23] is utilized as the XAI technique to explain individual predictions by revealing the most influential features enhancing DDoS attack detection. The obtained explanations furnish insights into the model's prediction mechanism, supporting better explainability with increasing trust in the federated DDoS attack-detection system.

E. Experimental Configuration

The federated training process was conducted using four clients over ten communication rounds, with each client performing three local training epochs per round. Following the stratified 80/20 train-test split, the training data were distributed among the four clients, with each client retaining its assigned data locally throughout the federated training process. The client data were partitioned before training to enable collaborative learning without sharing raw network traffic with the central server. The models were trained using the Adam optimizer with a learning rate of 0.001 and a batch size of 16. For FedProx, the proximal coefficient was set to 0.01. Binary cross-entropy was used as the loss function.

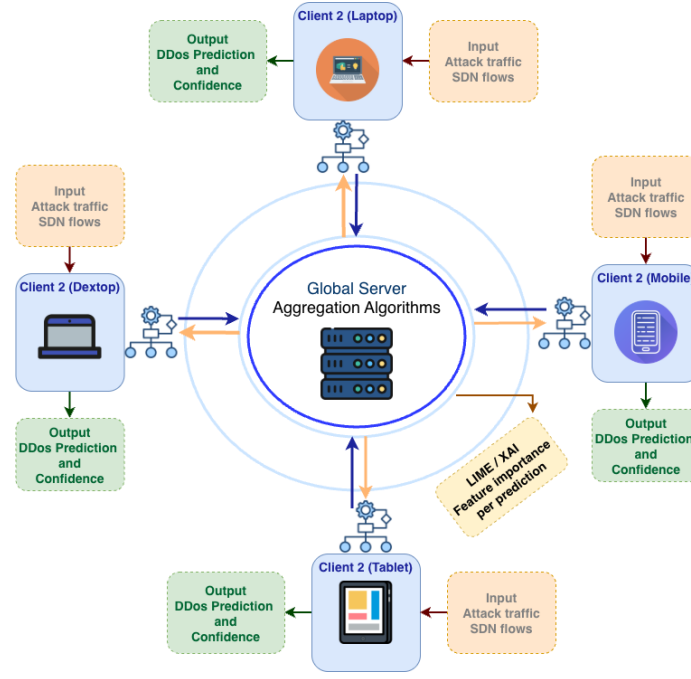


Fig. 1. Architecture of the proposed FedXDDoS framework for FL-based DDoS attack detection and LIME-based explainability.

IV. RESULTS AND DISCUSSION

The experimental results are presented through classification performance, convergence analysis under FedAvg and FedProx, and LIME-based explainability.

A. Performance Evaluation Metrics

The performance of FedXDDoS is evaluated using accuracy, precision, recall, F1-score, and binary cross-entropy (BCE) loss. Accuracy measures overall classification correctness, while precision, recall, and F1-score evaluate the detection of attack traffic. BCE loss measures the difference between predicted probabilities and the actual labels, with lower values indicating better optimization.

B. Experimental Results

1) *Classification Performance*: Table II presents the performance of the five federated deep learning models. The MLP achieved the highest performance, with 99.69% accuracy and a 99.67% F1-score, indicating that it effectively captured discriminative patterns in the SDN traffic features. BiLSTM achieved the second-highest accuracy of 99.33%, while LSTM and 1D-CNN achieved 99.05%. GRU obtained the lowest accuracy at 98.80%. Despite these differences, all models exceeded 98% accuracy, indicating consistent classification performance across the evaluated architectures.

2) *FedAvg Convergence Analysis*: Fig. 2(a)–(b) shows the accuracy and loss convergence under FedAvg. All models improved across communication rounds, indicating effective collaborative learning. MLP converged faster and achieved the highest final accuracy, whereas GRU showed comparatively

TABLE II
PERFORMANCE COMPARISON OF FEDERATED DL MODELS

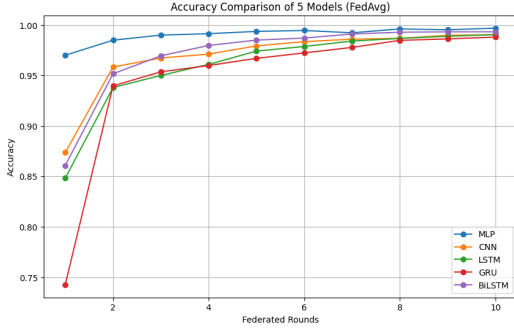
Model	Precision	Recall	F1-score	Accuracy
MLP	0.9968	0.9966	0.9967	0.9969
LSTM	0.9903	0.9896	0.9900	0.9905
1D-CNN	0.9899	0.9901	0.9900	0.9905
BiLSTM	0.9939	0.9920	0.9930	0.9933
GRU	0.9873	0.9875	0.9874	0.9880

slower convergence. The loss curves also decreased progressively, with MLP achieving the lowest final loss, consistent with its superior classification performance.

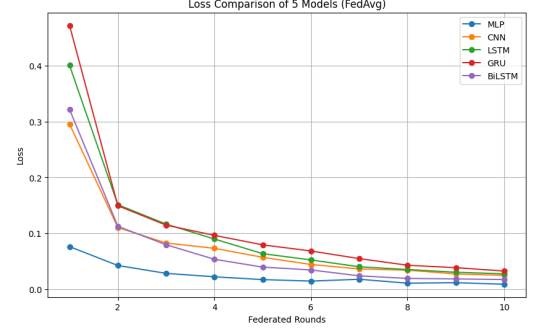
3) *FedProx Convergence Analysis*: Fig. 2(c)–(d) presents the convergence behavior under FedProx. The proximal regularization provides more constrained local updates, helping reduce client drift in heterogeneous settings. The loss values in Fig. 2(d) decreased progressively, while MLP again achieved the highest accuracy and lowest loss among the evaluated architectures.

C. Explainability Analysis Using LIME

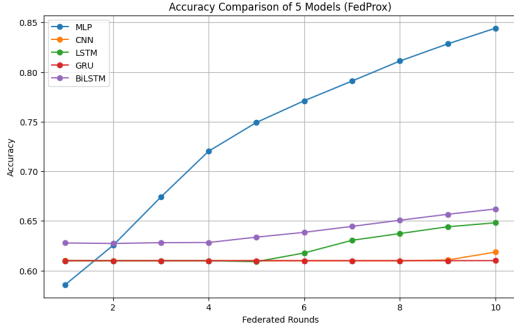
LIME was used to provide feature-level interpretations of the federated MLP predictions. Fig. 3(a)–(d) shows explanations for normal traffic instances, while Fig. 3(e)–(f) presents explanations for DDoS traffic instances. Across the analyzed samples, *pktrate*, *pktcount*, *pktperflow*, and *Protocol* repeatedly appeared among the influential features. Their relative contributions varied across individual instances, reflecting the instance-level nature of LIME. These explanations provide qualitative insight into the traffic characteristics influencing



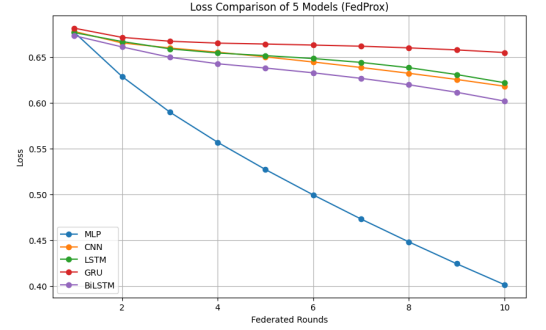
(a) FedAvg accuracy.



(b) FedAvg loss.



(c) FedProx accuracy.



(d) FedProx loss.

Fig. 2. Accuracy and loss convergence under FedAvg and FedProx.

the federated MLP predictions for both benign and malicious traffic.

V. CONCLUSION

This study presented FedXDDoS, an explainable FL framework for privacy-preserving DDoS attack detection in SDN environments. The proposed framework integrates MLP, 1D-CNN, LSTM, GRU, and BiLSTM architectures under both FedAvg and FedProx optimization strategies to enable collaborative learning without exposing raw network traffic data. The results showed that the framework achieved strong detection performance. Using FedAvg with MLP, the framework achieved an accuracy of 99.69% and an F1-score of 99.67%. In addition, all evaluated models attained accuracies above 98%, demonstrating the effectiveness of the framework. LIME-based explainability improved model transparency by identifying the

most influential traffic features, while comparative analysis showed that FedAvg achieved higher overall performance and FedProx provided improved training stability under heterogeneous environments.

However, the experimental evaluation was conducted using only the DDoS-SDN dataset, which may limit the generalizability of the findings across different traffic distributions and network environments. In addition, the current LIME analysis provides qualitative instance-level explanations rather than a quantitative evaluation of explanation faithfulness and stability. Future work will therefore evaluate FedXDDoS on additional publicly available DDoS and intrusion-detection datasets and incorporate quantitative measures of explanation faithfulness and stability to further assess its robustness, generalization capability, and interpretability.

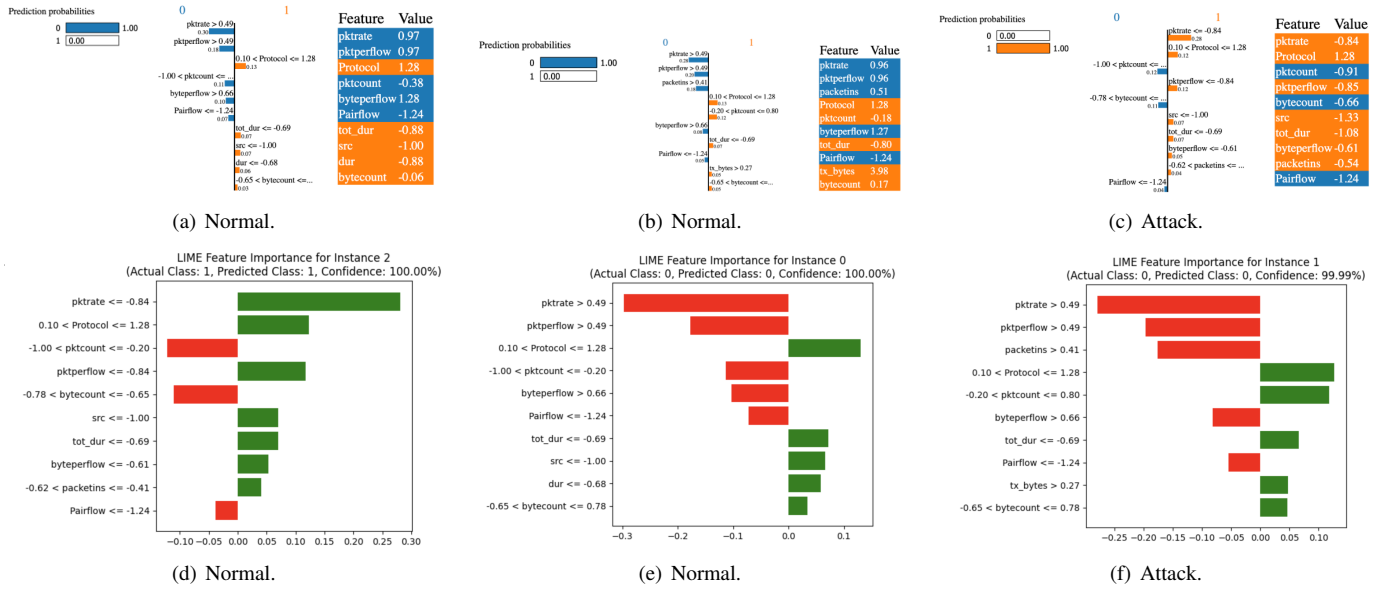


Fig. 3. Representative LIME explanations for normal and DDoS traffic predictions.

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